12TH CHEMISTRY · CHAPTER 23 · CHAPTER TEST

Carbonyl Compounds and Carboxylic Acids

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 23	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Oxidation of a secondary alcohol gives:

(a) Aldehyde (b) Carboxylic acid
(c) Ester (d) Ketone

2. Tollens' reagent with an aldehyde gives:

(a) Yellow precipitate (b) Silver mirror
(c) Brick-red precipitate (d) Orange crystals

3. Which compound gives the iodoform test?

(a) Propanone (b) Methanal
(c) Pentan-3-one (d) Methanol

4. Carboxylic acids are reduced to primary alcohols by:

(a) H_2/Ni (b) $NaBH_4$
(c) $LiAlH_4$ (d) $KMnO_4$

5. The strongest acid is:

(a) Acetic acid (b) Trichloroacetic acid
(c) Chloroacetic acid (d) Ethanol

6. Which is hydrolysed most easily?

(a) Chloroethane (b) Chlorobenzene
(c) Ethanoyl chloride (d) Ethane

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Why is ethanal distilled off as soon as it is formed?
ii. Why are aldehydes more reactive than ketones?
iii. What is Fehling's solution test?
iv. What is glacial acetic acid?
v. Why is methanoic acid a strong reducing agent?
vi. Why are the boiling points of esters lower than those of alcohols?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain the mechanism of addition of HCN to carbonyl compounds.
(b) Describe four methods for the preparation of carboxylic acids.

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